Murat Ambarkutuk

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Professional Experience

R^2 Robotics - RANPAK **November 2024 – Present**
Principal Perception Engineer *January 2025 – Present*
Sr. Perception Engineer *November 2024 – January 2025*

- Own the machine learning and perception stack end to end, from development through runtime, across Linux amd64 and arm64 platforms, including NVIDIA Jetson Nano, Orin Nano, and NX (CUDA, TensorRT), deployed on **[fleet-devices]** devices across **[fleet-sites]** sites.
  - Created r2-bb, the training path: pretrained ViT and ConvNeXt V2 backbones in-house on **[pretrain-images]** images, froze them, and trained task heads on them.
  - Defined the downstream contracts that let independent task heads solve many perception problems from one backbone pass, amortizing inference and enabling edge-level optimizations at industrial scale, cutting p50 latency from **[amort-sep-ms]** ms to **[amort-shared-ms]** ms for **[amort-k]** heads on Orin Nano, with **[amort-mem-pct]**% less memory and **[amort-energy-pct]**% less energy per frame.
  - Built task heads on the frozen backbone: a fine-tuned, deployed multi-head classifier, a heatmap-based keypoint estimator, and segmentation and oriented-box heads, **[heads-prod]** in production and **[heads-dev]** in development; **[cls-f1]** classifier F1, **[kp-oks]** keypoint OKS, **[seg-miou]** segmentation mIoU.
  - Built libinference, the inference path: a header-only, cross-platform ONNX/TensorRT library with dynamic shapes, multi-GPU support, CUDA stream parallelism, and unified logging, with **[libinf-overhead]**% overhead over raw TensorRT and **[libinf-load-s]** s engine load on Orin Nano.
  - Compressed models with knowledge distillation and pruning for efficient inference (**[comp-size-x]**× smaller, **[comp-speed-x]**× faster, **[comp-acc-pt]** pt accuracy cost); in-house pretrained teacher and student, teacher swappable for any ViT- or ConvNeXt V2-based foundation vision encoder.
  - Built synthetic data generation pipelines with NVIDIA Isaac Sim and a Blender-based native generator, **[synth-images]**k images lifting held-out real accuracy **[synth-lift]** pt.
- Architect Decision Tower 2.0, a full rewrite of the second-generation perception stack for end-of-line packaging (quality control and order verification), on the same frozen backbone with amortized inference, sustaining **[dt-ipm]** items per minute per station.
  - Built multi-camera calibration, multi-camera 3D reconstruction, and neural scene representations (**[dt-reproj]** px reprojection error, **[dt-recon-mm]** mm reconstruction error).
- Build and maintain the MLOps stack for the ML lifecycle: MLflow (PyTorch, TensorFlow, Keras), a private S3-backed APT repository with release and dev channels, and the full maintenance and release ceremonies of libinference, shipping **[rel-per-month]** releases a month across **[apt-channels]** channels.
- Own the project board: divide products into MVPs (**[mvps-shipped]** shipped), develop technical plans, and execute them with agile and a demo-every-week mentality; agentic development and review pipelines let the six-person team operate ten-strong, merging **[velocity-x]**× more changes per week.
- Host cross-functional engineering meetings to align with the product roadmap, and mentor team members on computer vision, deep learning optimization, and production deployment.

Baykar Defense (TR) - Atlas Imaging (US) **2018 – 2020**
Computer Vision Engineer *Remote*

- Developed orthorectified maps from aerial photos taken by autonomous UAVs. Derived the pose (location and orientation) of the UAV and its components from distributed Inertial Navigation System measurements, then used it in an iterative computer vision algorithm that projects image corners onto the ground while accounting for elevation changes in the scene.
- Built a sensor fusion algorithm, with a statistical representation, to correct the results of the earlier stages of the map-making pipeline.
- Introduced test-driven development to the team: wrote tests for coverage, style requirements, and the performance and accuracy of the algorithms.
- Architected the CI/CD pipeline that runs these tests automatically and generates reports from the results.

Education

PhD in Computer Engineering **2018 – 2024**
The Bradley Department of Electrical and Computer Engineering at Virginia Tech *Blacksburg, VA*

- Paul E. Torgersen Research Excellence Award

- The Outstanding Research Award and Fellowship from the Turkish Ministry of National Education

Dissertation: A Sensor Fusion Technique for Gait Analysis from Structural Vibration and Computer Vision

MS in Mechanical Engineering **2015 – 2018**

Mechanical Engineering at Virginia Tech *Blacksburg, VA*

Thesis: A Grid based Indoor Radiolocation Technique Based on Spatially Coherent Path Loss Model

BS in Mechatronics Engineering **2008 – 2013**

Mechatronics Engineering Department at Kocaeli University *Kocaeli, Turkey*

Thesis: Box Dimensioning System with Kinect Sensor

Technical & Language Skills

Languages: Python, C++, C, CUDA, MATLAB
ML/CV: PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, scikit-image, YOLO, DINOv3, ViT, ConvNeXt V2
3D Vision & Robotics: COLMAP, Multi-view Geometry, TSDF Fusion, 3D Gaussian Splatting, ROS
Simulation & NVIDIA Stack: Isaac Sim (Omniverse), Jetson, Blender, Synthetic Data Generation, CUDA, TensorRT
Deployment, Optimization & MLOps: ONNX, MLflow, CUDA Kernels, Quantization, Pruning, Model Distillation
Tools & Technologies: Linux (amd64, arm64), APT, AWS S3, Git, GitHub, GitLab, CI/CD, Docker, LaTeX
Language Proficiency: Turkish (Native), English (Proficient), German (Novice)

Projects

İşaret: Agent-Run AI & Robotics Newsletter

oruntu | isaret.ai **May 2026 – Present**

- Built and operate an AI-agent pipeline that curates arXiv papers and industry news, writes Turkish summaries with source links and license-credited figures, runs automated review, and publishes each issue; 50+ issues since May 2026.
- Deliver every issue by email, Atom/RSS, a JSON API, LinkedIn, and Telegram; Cloudflare-hosted, GitHub-published.

Sensor Fusion for Occupant Localization in Smart Buildings

Advisors: P Plassmann, CF Jones | Related publications: [1–5] **2018–2024**

- Developed a robust algorithm that estimates heel-strike locations from the accelerometer measurements of a vibrating floor.
- Built a computer vision algorithm that localizes heel-strike locations from stereoscopic camera images.
- Designed a sensor fusion framework that combines the vibration- and vision-based heel-strike locations.
- Created a combined simulation environment that generates the vibration and visual data needed to validate the algorithms.
- Quantified the efficacy and accuracy of the implementations with simulations and controlled experiments.

A Grid-based Indoor Radiolocation Technique based on Spatially Coherent Path Loss Model

Advisor: T Furukawa | Publications: [6] **2016–2017**

- Derived a probabilistic framework for passive radio localization under multipath propagation that determines the location of occupants, robots, and inventory in indoor environments.
- Developed the localization algorithm based on the probabilistic framework.
- Validated the algorithm with controlled experiments whose results were presented at an IEEE RAS conference.

Relevant Publications

[1] **M. Ambarkutuk** and P. E. Plassmann. Modeling and analysis of dispersive propagation of structural waves for vibro-localization. *Sensors*, 24(23):7744, 2024. URL https://doi.org/10.3390/s24237744.

[2] **M. Ambarkutuk**, S. Alajlouni, P. A. Tarazaga, and P. E. Plassmann. A multi-sensor stochastic energy-based vibro-localization technique with byzantine sensor elimination. *Sensors*, 23(23):9309, 2023. URL https://doi.org/10.3390/s23239309.

[3] **M. Ambarkutuk** and P. Tarazaga. Uncertainty analysis of an occupant localization technique based on simulated structural vibrations. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 39th IMAC*. Springer, 2022.

[4] S. Alajlouni, V. V. S. Malladi, and **M. Ambarkutuk**. Investigation of using log-spectrum averaging (cepstral averaging) for blind reconstruction of an unknown impact input force. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 39th IMAC*. Springer, 2022.

[5] S. Alajlouni, **M. Ambarkutuk**, and P. Tarazaga. Towards the detection and localization of multiple occupant footsteps from vibroacoustic measurements. In *Dynamics of Civil Structures, Volume 2: Proceedings of the 38th IMAC, A Conference and Exposition on Structural Dynamics 2020*, page 193. Springer, 2020.

[6] **M. Ambarkutuk** and T. Furukawa. A grid-based indoor radiolocation technique based on spatially coherent path loss model. In *2017 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, pages 220–226. IEEE, 2017.

[7] H. Ocak, **M. Ambarkütük**, G. Küçükyıldız, and S. Karakaya. Image processing based package volume detection with kinect. In *2015 23nd Signal Processing and Communications Applications Conference (SIU)*, pages 515–518. IEEE, 2015.

Community Service & Leadership

Graduate Student Ambassador, ECE at Virginia Tech · Student Member & Reviewer, IEEE Robotics and Automation Society · Student Member, Society of Experimental Mechanics · Secretary, Turkish Student Association at Virginia Tech